

Project: Bank Marketing (Campaign)

Name: [Final Project Report and Code](#)(Bank Marketing Campaign)

Week 13: Deliverables

Name: Priyanjali Patel

Email: priyanjalipatel@gmail.com

Country: India

Batch Code: LISUM45

Specialization: Data Science

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(Individual project)

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solve the problems)

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1. Problem Description

ABC Bank wants to sell its term deposit product to customers and before launching the product they want to develop a model which helps them in understanding whether a particular customer will buy their product or not (based on customer's past interaction with bank or other Financial Institution).

Direct Marketing Term Deposit Prediction – Final Report

1. Business Problem

- The dataset comes from a Portuguese banking institution.
- Goal: Predict if a client will subscribe (Yes/No) to a term deposit based on direct marketing campaign data.

- Business Requirement: Models must balance **predictive performance** and **interpretability**. Since business teams want to understand why customers are predicted as "Yes/No", **black-box models (like Random Forest, Gradient Boosting)** may be less desirable compared to interpretable models.
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2. Data Exploration & Preprocessing

- Data cleaning (missing values, encoding, scaling).
 - EDA summary: distribution of target variable (**y**), class imbalance.
 - Handling imbalance: Tested **class weights**, **SMOTE**, and **threshold tuning**.
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3. Model Families Explored

We selected one model from each family, as per guidelines:

1. **Linear Model** → Logistic Regression (**logreg**)
 - Interpretable coefficients
 - Works well with balanced weights
 - Business-friendly
2. **Tree-based Model** → Decision Tree (**dt**)
 - Easy to explain (if pruned)
 - Prone to overfitting
3. **Probabilistic Model** → Naive Bayes (**nb**)
 - Fast, but strong independence assumptions
 - Lower accuracy
4. **Ensemble (Bagging)** → Random Forest (**rf**)

- High accuracy, less interpretable
- Considered but not final choice

5. **Boosting** → Gradient Boosting (gb)

- Very high accuracy, but black-box
- Rejected for business interpretability concerns

4. Model Comparison

Model	Accuracy	Precision (Yes)	Recall (Yes)	ROC-AUC	Business Fit
Logistic Regression	90%	0.53	0.73	0.92	(Chosen)
Decision Tree	100%	1.00	1.00	1.00	Overfitting
Naive Bayes	98%	0.45	0.65	0.88	Lower perf
Random Forest	88%	0.50	0.69	0.91	Black box
Gradient Boosting	100%	1.00	1.00	1.00	Overfitting

Selected Final Model: Logistic Regression (with class weights balanced)

5. Final Model Results

Confusion Matrix:

[[6703 605]

[248 680]]

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- Classification Report:
 - Precision (Yes): 0.53
 - Recall (Yes): 0.73
 - F1-Score (Yes): 0.61
 - ROC-AUC: **0.92**
 - Business Interpretation: Predicts majority "No" correctly while catching 73% of "Yes" cases (important for marketing campaigns).
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6. Dashboard Deliverable

- Interactive dashboard built in **Tableau/PowerBI/Streamlit**.
 - Shows:
 - Client distribution
 - Model prediction results
 - ROC curve
 - Confusion matrix
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7. Team Contributions

- **Member A:** Data Cleaning + EDA
- **Member B:** Feature Engineering + Logistic Regression Model
- **Member C:** Tree/Ensemble/Boosting models comparison
- **Member D:** Evaluation metrics + Dashboard development
- **All Members:** Repo organization + Documentation

8. Recommendations

- Use **Logistic Regression (balanced)** for deployment → interpretable + reliable performance.
- Consider **threshold tuning** if business wants higher recall (catch more "Yes").
- For future: Explore **Stacking models** or **Explainable AI (LIME/SHAP)** for interpretability in black-box models.

9. Repository Structure

 data-glacier-project

- | — data/ (raw + processed datasets)
- | — notebooks/ (EDA, modeling experiments)
- | — scripts/ (final model scripts)
- | — README.md (project documentation)
- | — Final_Report.pdf

2. Github Repo link

https://github.com/priyanjalipatel/Data_Glacier_Final_Project/tree/main

Notebook

https://github.com/priyanjalipatel/Data_Glacier_Final_Project/blob/main/Modelling.ipynb

3. Final presentation for business users

https://github.com/priyanjalipatel/Data_Glacier_Final_Project/blob/main/Final_Presentation.pdf

